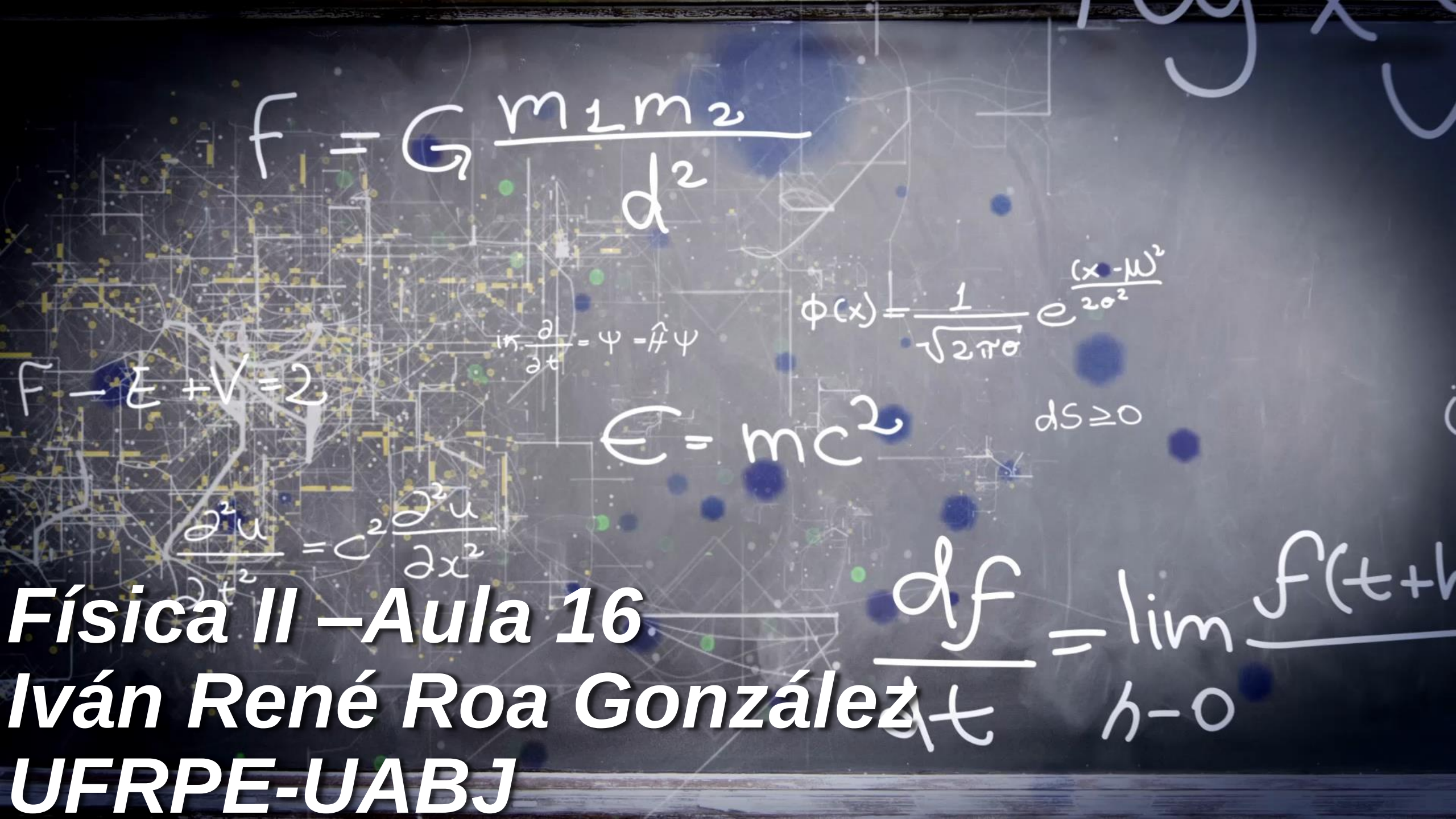


Física II –Aula 16

Iván René Roa González

UFRPE-UABJ

$$x(t) = c_1 e^{\left(\frac{-b+\sqrt{b^2-4km}}{2m}\right)t} + c_2 e^{\left(\frac{-b-\sqrt{b^2-4km}}{2m}\right)t}$$

criticamente amortecido $\longrightarrow$ $b = \sqrt{4\ \mathrm{km}}$

$\sqrt{b^2 - 4km} = 0$

$$\boxed{b_{crit} = 2\omega_0 m}$$

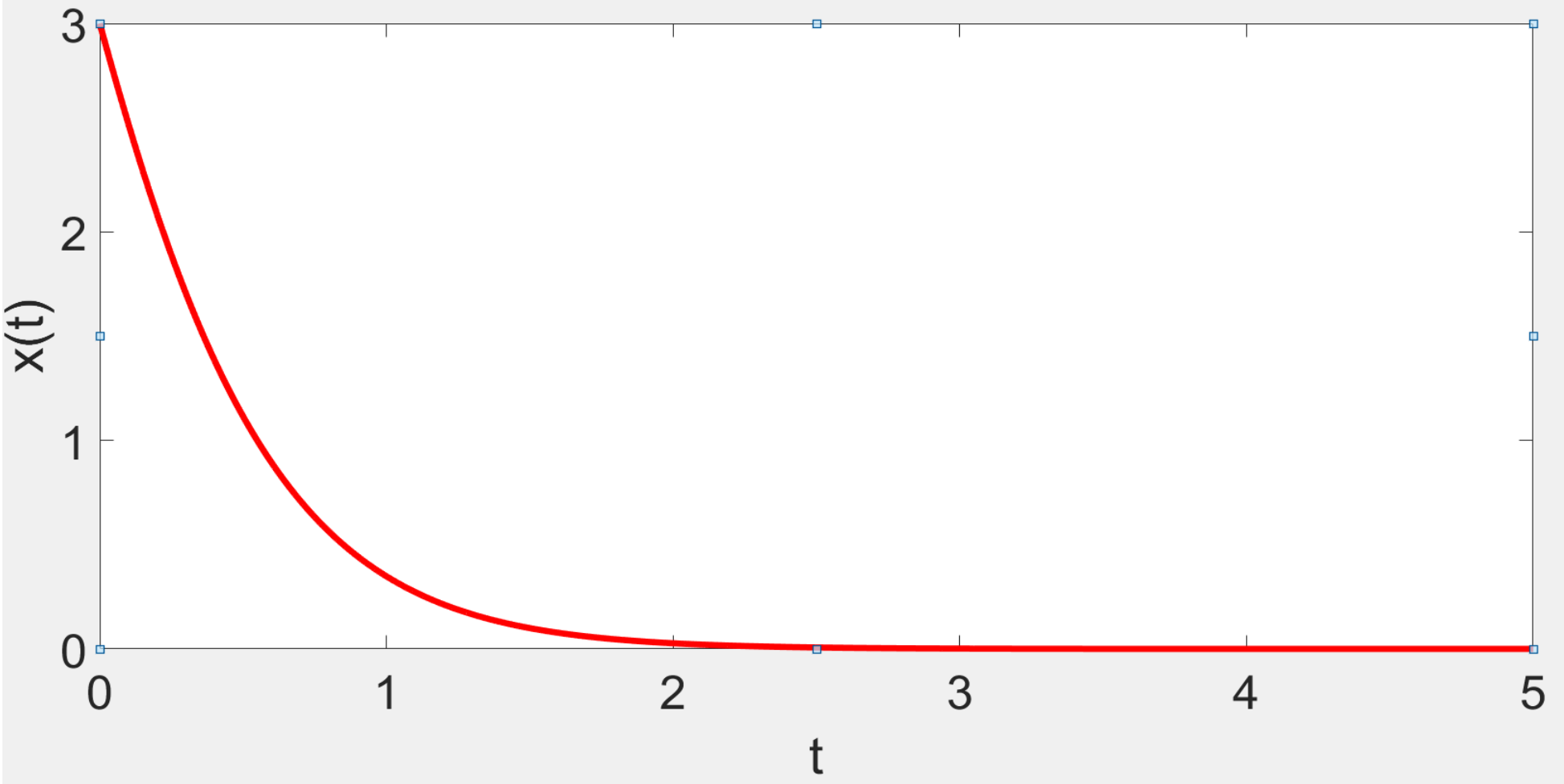
$\longleftarrow$ $\sqrt{k} = \omega_0\sqrt{m}$

$$x(t) = c_1 e^{\frac{-b}{2m}t} + c_2 e^{\frac{-b}{2m}t} = \underbrace{(c_1 + c_2)}_{a_1} e^{\frac{-b}{2m}t}$$

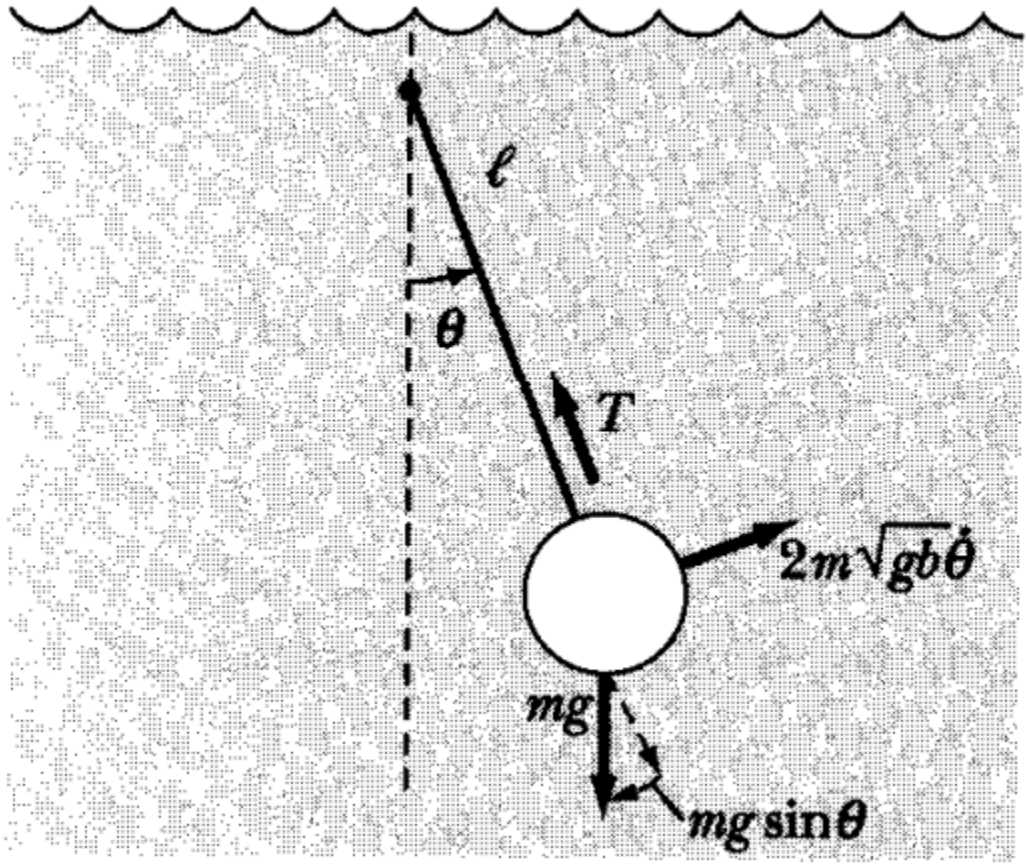
$$x(t) = a_1 e^{\frac{-b}{2m}t}$$

- Pode-se verificar que $a_2 t e^{-\frac{b}{2m}t}$ também é uma solução, portanto a solução geral será uma combinação destes dois:

$$x(t) = (a_1 + a_2 t)\, e^{-\frac{b}{2m}t}$$



Exemplo



$$F_{\text{res}} = 2m\sqrt{g/\ell}\,(\ell\dot{\theta})$$

$t = 0$ $\quad \theta = \alpha \qquad \dot{\theta} = 0$

$$m\ell\ddot{\theta} = -mg\sin\theta - 2m\sqrt{g/\ell}(\ell\dot{\theta})$$

$$\sin\theta \approx \theta$$

$$\ddot{\theta} + 2\sqrt{g/\ell}\,\dot{\theta} + \frac{g}{\ell}\theta = 0$$

$$\omega_0^2 = g/\ell$$

criticamente amortecido

$$\xi\omega_0 = \beta$$

$$\omega_0^2 = \beta^2 \quad \Rightarrow \quad \beta^2 = g/\ell$$

$$\theta(t) = (A + Bt)\,e^{-\beta t}$$

$$\theta(t=0) = \alpha = A$$

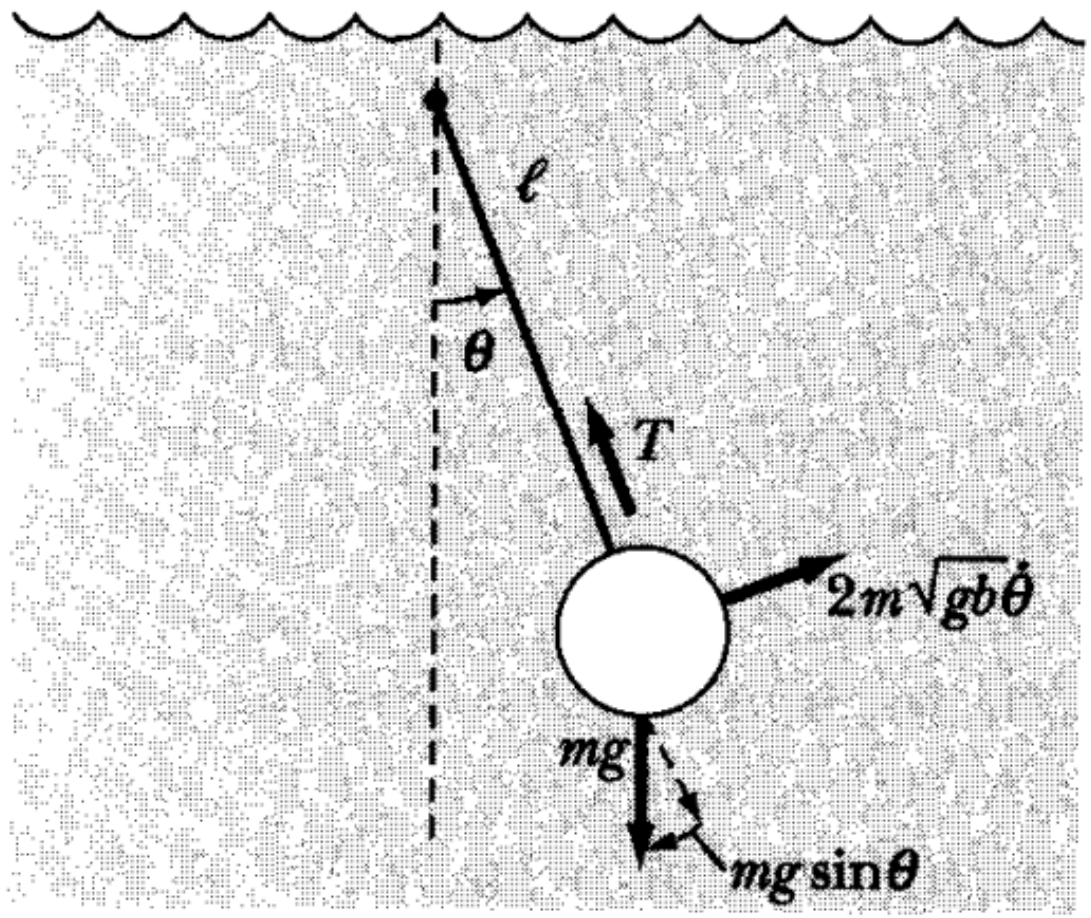
$$\dot{\theta}(t) = Be^{-\beta t} - \beta(A + Bt)\,e^{-\beta t}$$

$$\dot{\theta}(t=0) = 0 = B - \beta A$$

$$B = \beta A = \beta\alpha$$

$$\theta(t) = \alpha\left(1 + \sqrt{g/\ell}\;t\right)e^{-\sqrt{g/\ell}\,t}$$

$$\dot{\theta}(t) = \frac{-\alpha g}{\ell}\,t\,e^{-\sqrt{g/\ell}\,t}$$



VARIAÇÃO DA AMPLITUDE NO OSCILADOR SUBAMORTECIDO

$$ma = -kx - bv$$

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + b\frac{\mathrm{d}x}{\mathrm{d}t} + kx = 0. \longrightarrow \omega_o^2 = k/m, \quad \gamma = b/m.$$

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \gamma\frac{\mathrm{d}x}{\mathrm{d}t} + \omega_o^2 x = 0.$$

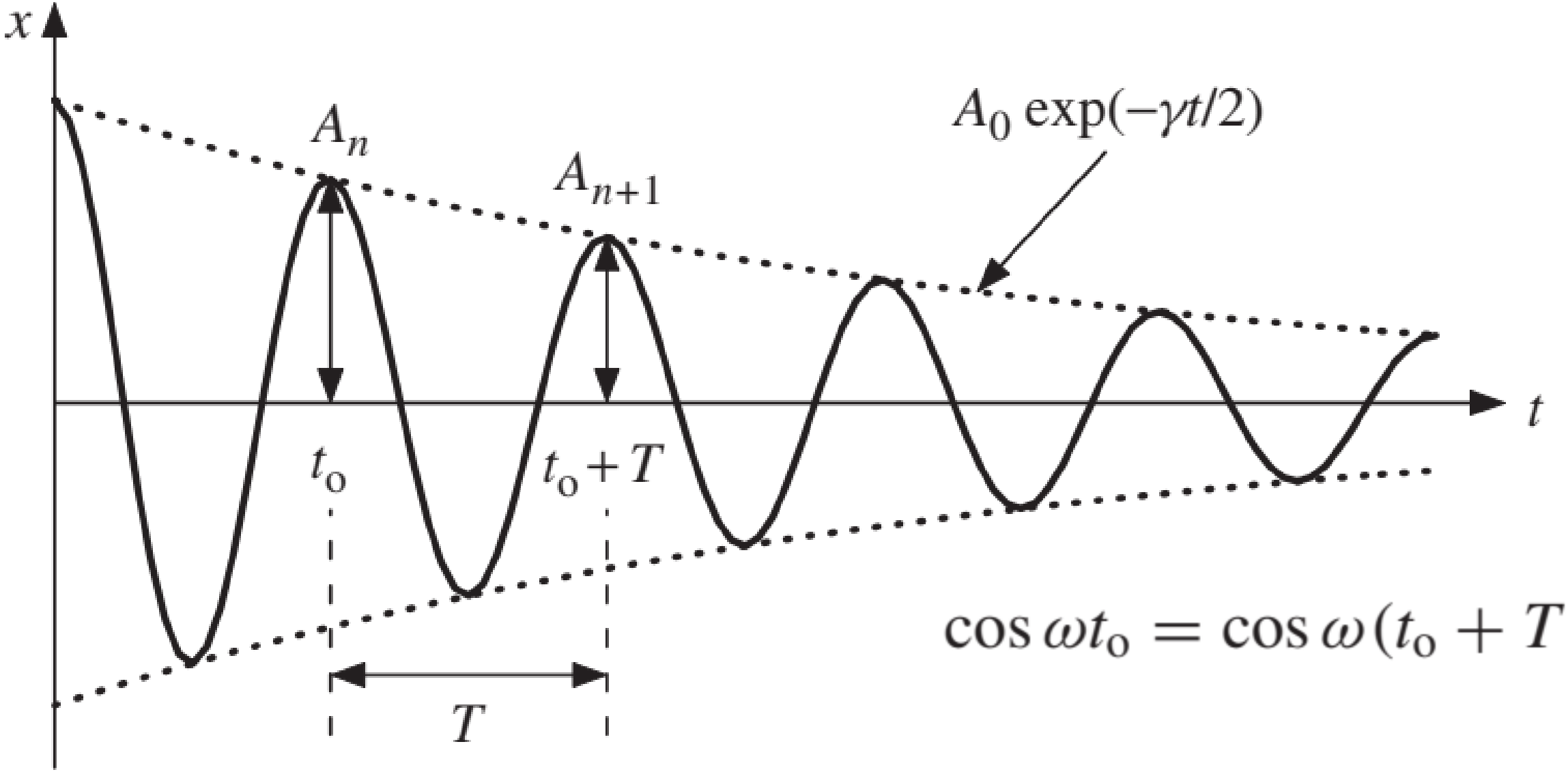
$$\boxed{\xi\omega_0 = \beta = \gamma/2}$$

subamortecido

$$\xi\omega_0 = \beta = \gamma/2$$

$$x = A_0 \exp(-\gamma t/2)\cos(\omega t + \phi) \longrightarrow \omega = (\omega_o^2 - \gamma^2/4)^{1/2}$$

$$\left.\begin{aligned} A_n &= x(t_o) = A_0 \exp(-\gamma t_o/2)\cos\omega t_o \\ A_{n+1} &= x(t_o + T) = A_0 \exp[-\gamma(t_o + T)/2]\cos\omega(t_o + T) \end{aligned}\right\} \phi = 0$$

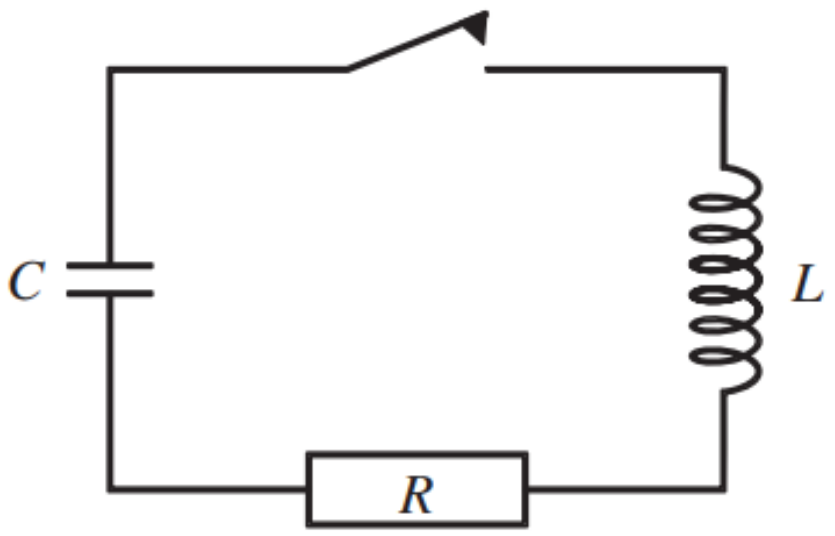


$$\cos \omega t_o = \cos \omega (t_o + T)$$

$$\frac{A_n}{A_{n+1}} = \exp\left(\frac{\gamma T}{2}\right) \Rightarrow \ln\left(\frac{A_n}{A_{n+1}}\right) = \frac{\gamma T}{2}$$

OSCILAÇÕES ELÉTRICAS AMORTECEDADAS

Mechanical		Electrical	
x	Displacement	q	Charge
$\dot{x}$	Velocity	$\dot{q} = I$	Current
m	Mass	L	Inductance
b	Damping resistance	R	Resistance
$1/k$	Mechanical compliance	C	Capacitance
F	Amplitude of impressed force	$\mathcal{E}$	Amplitude of impressed emf



$$L\frac{\mathrm{d}I}{\mathrm{d}t} + RI + \frac{q}{C} = 0$$

$$L\frac{\mathrm{d}^2q}{\mathrm{d}t^2} + R\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{q}{C} = 0.$$

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + b\frac{\mathrm{d}x}{\mathrm{d}t} + kx = 0.$$

$$q = q_0 \exp(-Rt/2L) \cos[(1/LC - R^2/4L^2)^{1/2} t]$$

Mechanical

$$\omega = \sqrt{\omega_0{}^2 - \left(\frac{b}{2m}\right)^2}$$

Electrical

$$\omega^2 = \frac{1}{LC} - \frac{R^2}{4L^2}$$

$$V_C = q/C = V_0 \exp(-Rt/2L) \cos[(1/LC - R^2/4L^2)^{1/2} t]$$

TAXA DE PERDA DE ENERGIA EM UM OSCILADOR HARMÔNICO

$$E = K + U = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

$$\gamma^2/4 \ll \omega_o^2 \qquad \omega \cong \omega_o$$

$$x = A_0 \exp(-\gamma t/2) \cos \omega_o t$$

$$v = \frac{\mathrm{d}x}{\mathrm{d}t} = -A_0\omega_o \exp(-\gamma t/2)[\sin\omega_o t + (\gamma/2\omega_o)\cos\omega_o t].$$

$$\gamma/2 \ll \omega_o$$

$$v = \frac{\mathrm{d}x}{\mathrm{d}t} = -A_0\omega_o \exp(-\gamma t/2)(\sin\omega_o t).$$

$$E = K + U = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

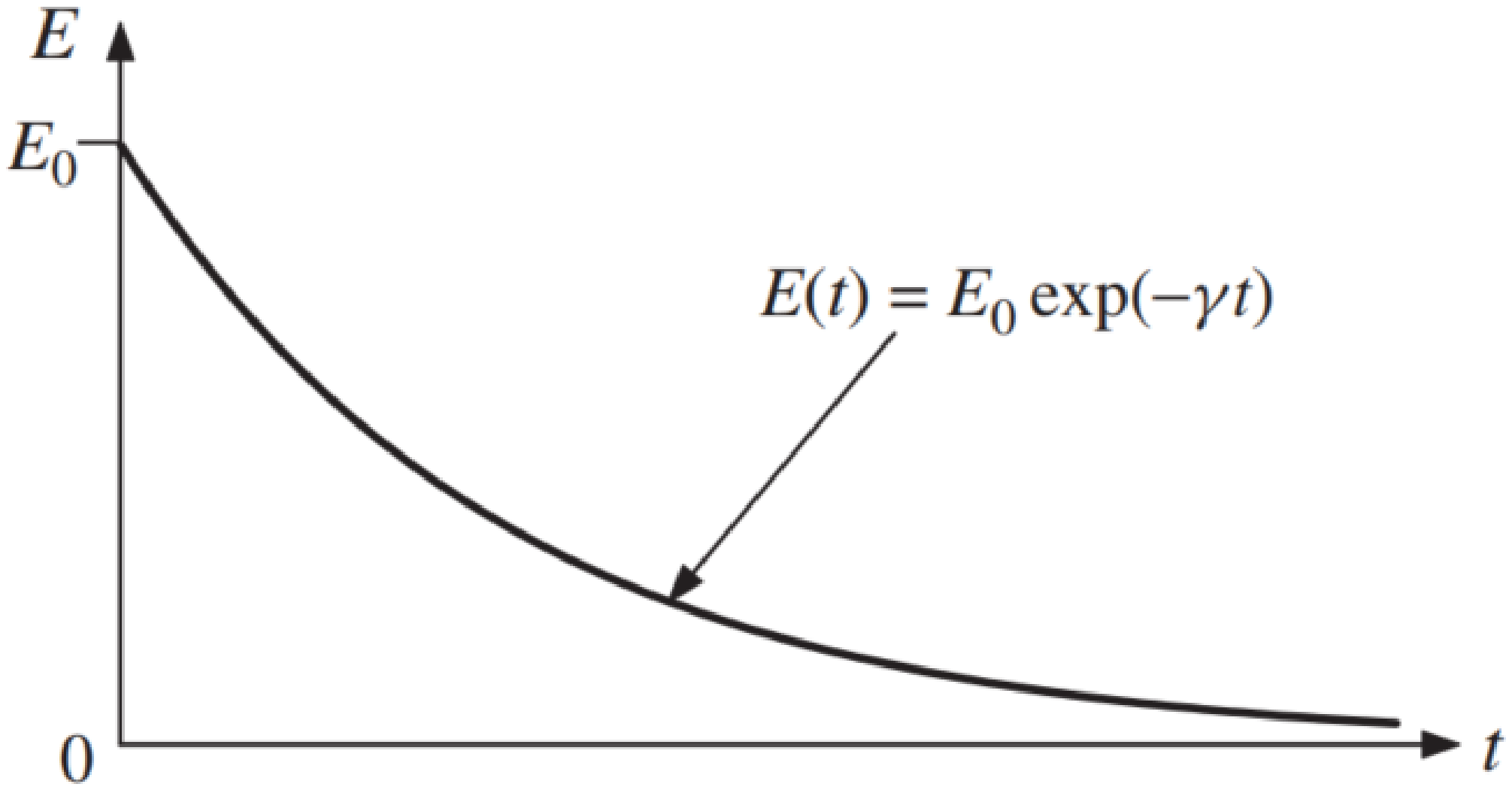
$$E = \frac{1}{2}A_0^2 \exp(-\gamma t)(m\omega_o^2 \sin^2\omega_o t + \underbrace{k}_{\omega_o^2 = k/m}\cos^2\omega_o t).$$

$$E = \underbrace{\frac{1}{2}kA_0^2}\exp(-\gamma t)$$

$$\boxed{E(t) = E_0 \exp(-\gamma t)}$$

$\tau = 1/\gamma$ é chamado de tempo de vida

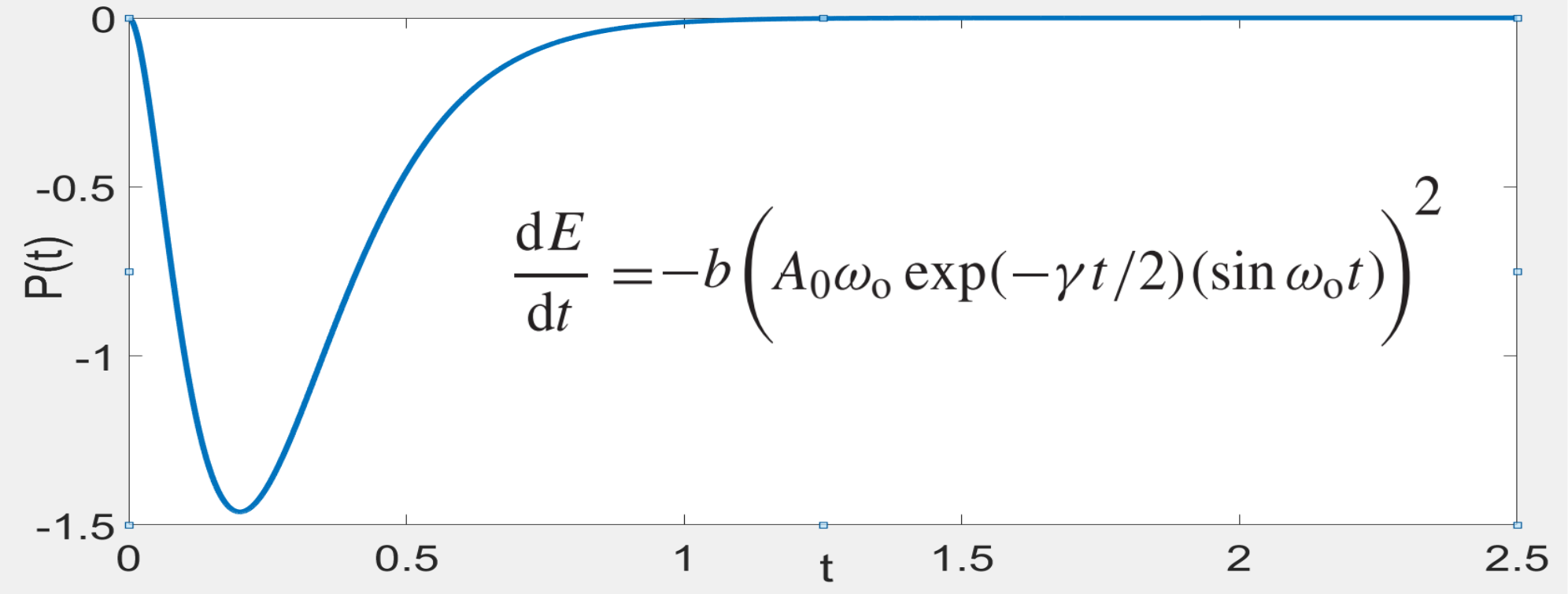
$$E(t) = E_0 \exp(-t/\tau)$$


E
E0
E(t) = E0 exp(−γt)
0
t

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}mv^2 + \frac{1}{2}kx^2\right) = mv\frac{\mathrm{d}v}{\mathrm{d}t} + kx\frac{\mathrm{d}x}{\mathrm{d}t} = (ma + kx)v$$

$$ma = -kx - bv$$

$$\frac{\mathrm{d}E}{\mathrm{d}t} = (-bv)v$$



$$\frac{\mathrm{d}E}{\mathrm{d}t} = -b\left(A_0\omega_o \exp(-\gamma t/2)(\sin\omega_o t)\right)^2$$

o fator de qualidade, **Q**:

$$Q = 2\pi\left[\frac{U_0(energia..armazenada)}{\Delta U(energia..perdida..por..ciclo)}\right] \Rightarrow \boxed{Q = \frac{\omega_0}{\gamma}.}$$

$$E_1 = E_0 \exp(-\gamma t), \qquad E_2 = E_0 \exp[-\gamma(t+T)]$$

$$\frac{E_2}{E_1} = \exp(-\gamma T). \Rightarrow e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \cdots.$$

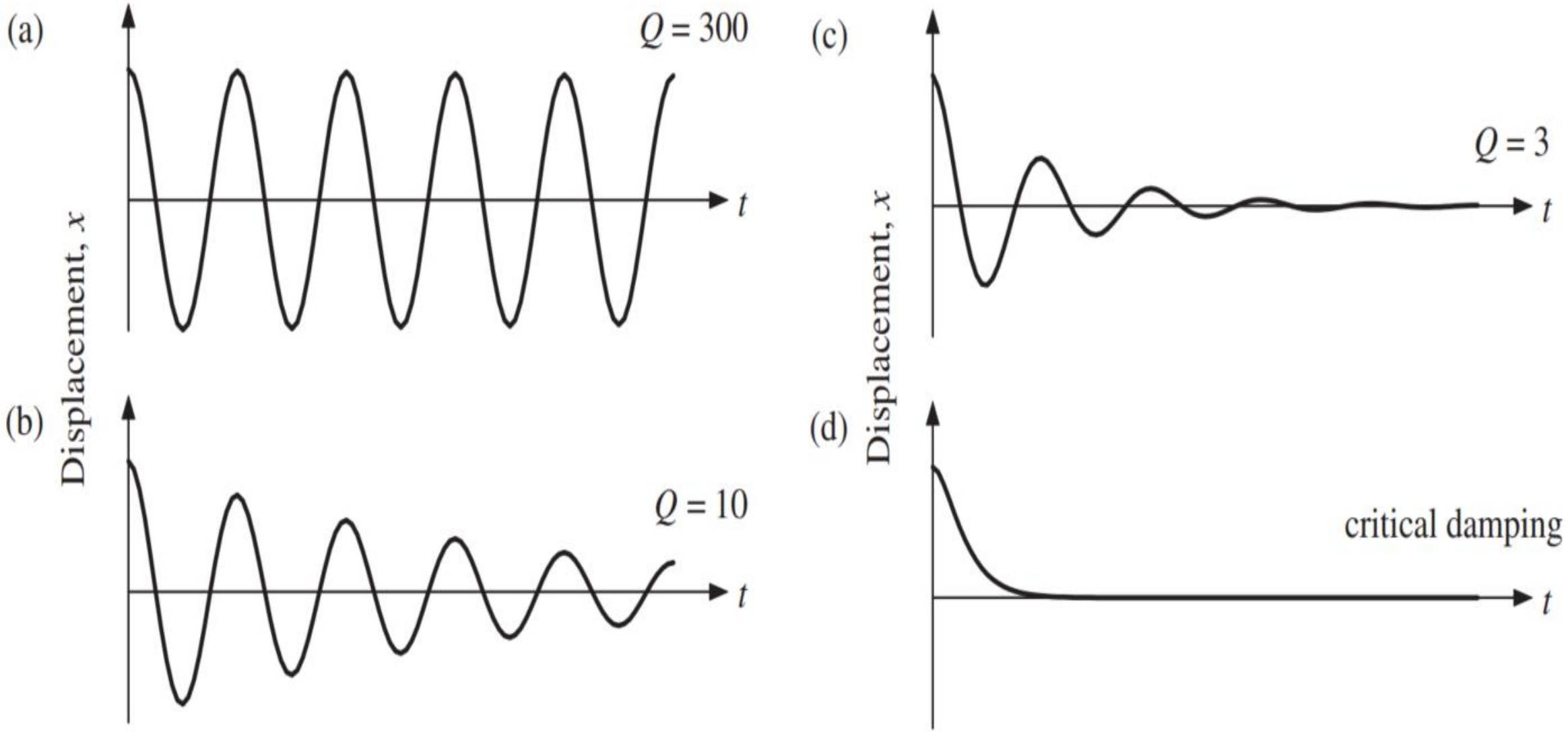
$$x \ll 1 \Rightarrow e^x \simeq 1 + x$$

$$\frac{E_2}{E_1} = \exp(-\gamma T). \longrightarrow e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \cdots.$$

$$x \ll 1 \longrightarrow e^x \simeq 1 + x$$

$$\frac{E_1 - E_2}{E_1} \simeq \gamma T \simeq \frac{2\pi\gamma}{\omega_o} = \frac{2\pi}{Q} \longrightarrow Q = \frac{\omega_0}{\gamma}$$

$$Q = 2\pi\left[\frac{U_0(energia..armazenada)}{\Delta U(energia..perdida..por..ciclo)}\right]$$


(a)
Q = 300
t
(b)
Q = 10
t
Displacement, x
(c)
Q = 3
t
(d)
critical damping
t
Displacement, x

Exemplo

- Quando a corda E de um violão (frequência 330 Hz) é tocada, a intensidade do som diminui por um fator de 2 após 4 s. Determine (i) o tempo de decaimento τ, (ii) o fator de qualidade Q e (iii) a perda de energia fracionada por ciclo.

(i)

$$E(t) = E_0 \exp(-t/\tau) \rightarrow \tau = \frac{t}{\ln[E_0/E(t)]} = \frac{4}{\ln 2} = 5.77 \text{ s.}$$

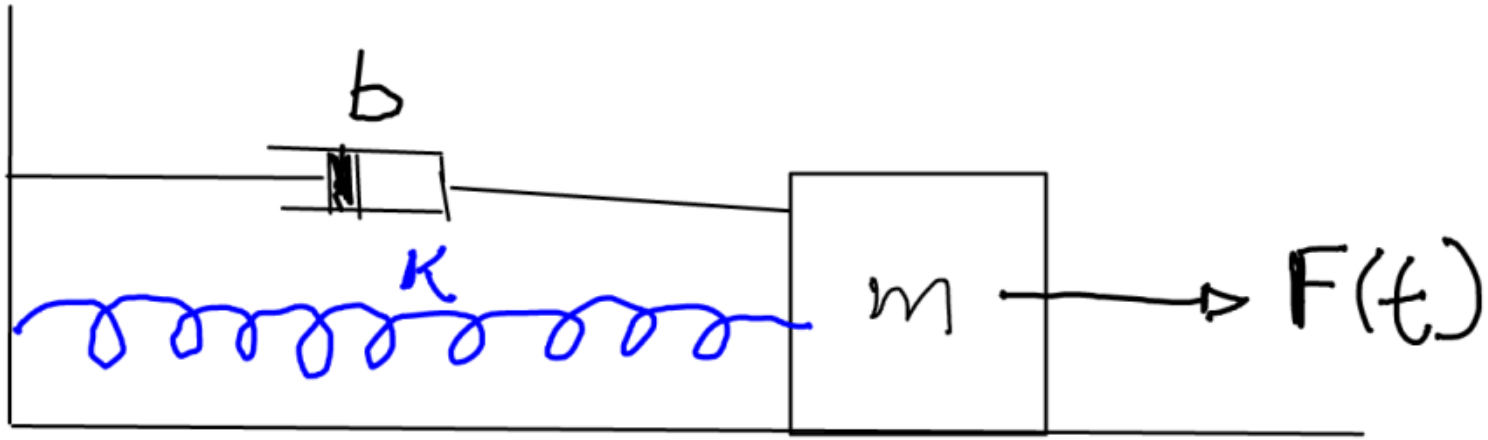
(ii)

$$Q = \omega_o/\gamma = \omega_o\tau = 2\pi \times 330 \times 5.77 = 1.2 \times 10^4.$$

(iii)

$$\frac{\Delta E}{E} = \frac{2\pi}{Q} = 5.3 \times 10^{-4}.$$

Oscilações Forçadas



2da lei de Newton

$$\sum \vec{F} = m\vec{a} \quad ; \quad \sum F_x = m a_x$$

$$-Kx - b\dot{x} + F(t) = m\ddot{x}$$

$$\boxed{F(t) = m\ddot{x} + b\dot{x} + Kx}$$

Eq. de movimento

$x_1(t)$ e $x_2(t)$ soluções da eq. de movimento

$$m\ddot{x}_2 + b\dot{x}_2 + kx_2 = F(t)$$

$$-\quad m\ddot{x}_1 + b\dot{x}_1 + kx_1 = F(t)$$

$$m(\ddot{x}_2 - \ddot{x}_1) + b(\dot{x}_2 - \dot{x}_1) + k(x_2 - x_1) = 0$$

$x_h(t) = x_2(t) - x_1(t)$: Solução homogênea

Solução geral $x(t) = x_h(t) + x_p(t)$

Nosso objetivo será determinar a solução particular $\boxed{x_p(t)}$

Casos de Forçador

(i) Forçador constante $F(t) = D = \text{constante}$

$$x_p = \frac{D}{K}$$

Justificativa $x_p = c$

$$\dot{x}_p = \emptyset$$

$$\ddot{x}_p = \emptyset$$

$$\boxed{F(t) = m\ddot{x} + b\dot{x} + Kx}$$

$$D = \emptyset + \emptyset + Kc$$

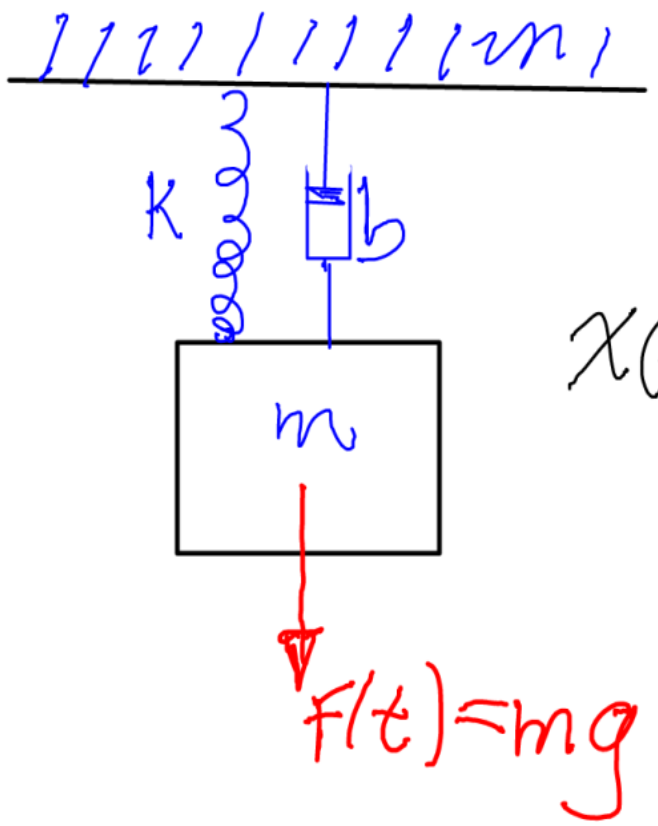
$$D/K = c$$

Uma oscilação subamortecida

$$x(t) = x_h(t) + x_p(t)$$

$$x(t) = A\exp\left(-\frac{b}{2m}t\right)\sin(\omega t + \varphi) + D/K$$

Exemplo



$$x(t) = A\exp\left(-\frac{b}{2m}t\right)\sin(\omega t + \varphi) + mg/K$$

Caso (ii) Forçador linear no tempo $F(t) = Et$

$x_p(t) = Bt + N$ B, N são constantes

$\dot{x}_p(t) = B$

$\ddot{x}_p(t) = 0$

$$F(t) = m\ddot{x} + b\dot{x} + kx$$

$$Et = 0 + bB + k(Bt + N)$$

$$Et = bB + kN + kBt$$

$$E = kB$$

$$\boxed{E/k = B}$$

$$0 = bB + kN$$

$$-b\left(\frac{E}{k}\right) = kN$$

$$\boxed{-\frac{bE}{k^2} = N}$$

$$x_p(t) = Bt + N = \frac{E}{k}t - \frac{bE}{k^2}$$

$$x(t) = x_h(t) + x_p(t) = A\exp\left(-\frac{b}{2m}t\right)\sin(\omega t + \varphi) + \frac{E}{k}t - \frac{bE}{k^2}$$

Caso (iii) Forçador harmônico $F(t)=q\sin\Omega t$

* $x_p(t) = \underline{A^*}\sin(\Omega t+\underline{\delta})$

* $\dot{x}_p(t) = A^*\Omega\cos(\Omega t+\delta)$

* $\ddot{x}_p(t) = -A^*\Omega^2\sin(\Omega t+\delta)$

$F(t) = m\ddot{x} + b\dot{x} + Kx$

$$q\sin(\Omega t) = -mA^*\Omega^2\sin(\Omega t+\delta) + bA^*\Omega\cos(\Omega t+\delta)$$
$$+ KA^*\sin(\Omega t+\delta)$$

$$\overset{*}{A}[K - m\Omega^2]\sin(\Omega t + \delta) + \overset{*}{A} b\Omega\cos(\Omega t + \delta) = q\sin\Omega t$$

$$[K - m\Omega^2]\sin(\Omega t + \delta) + b\Omega\cos(\Omega t + \delta) = \frac{q}{A^*}\sin\Omega t$$

- Para $t = 0$

$$[K - m\Omega^2]\sin\delta + b\Omega\cos\delta = 0$$

$$[K - m\Omega^2]\sin\delta = -b\Omega\cos\delta$$

$$tg\,\delta = \frac{-b\Omega}{K - m\Omega^2}$$

$$\tan\delta = \frac{-(b/m)\Omega}{k/m - \Omega^2} = -\frac{2\xi\omega_0\Omega}{\omega_0^2 - \Omega^2}$$

- Para $\Omega t + \delta = 0$ $\qquad \Omega t = -\delta$

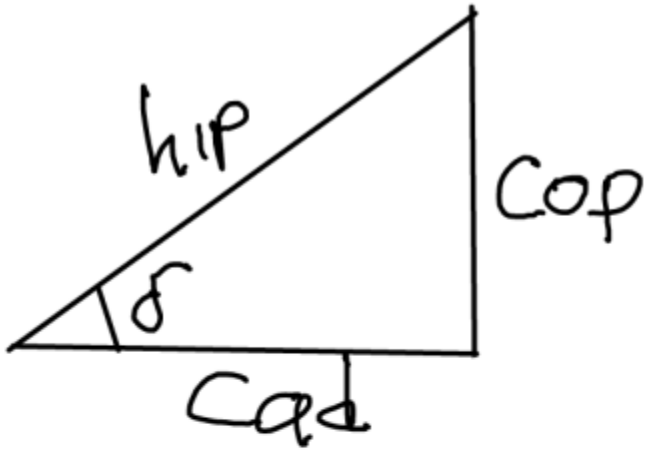
$$[K - m\Omega^2]\sin(\Omega t + \delta) + b\Omega\cos(\Omega t + \delta) = \frac{q}{A^*}\sin\Omega t$$

$$b\Omega = \frac{q}{A^*}\sin(-\delta)$$

$$A^* = \frac{-q}{b\Omega}\sin(\delta)$$

$$\tan\delta = -\frac{2\xi\omega_0\Omega}{\omega_0^2-\Omega^2}$$

$$\sin\delta = \frac{cop}{hip}$$



$$\tan\delta = \frac{cop}{cad}$$

$$\sin\delta = \frac{cop}{\sqrt{cop^2 + cad^2}}$$

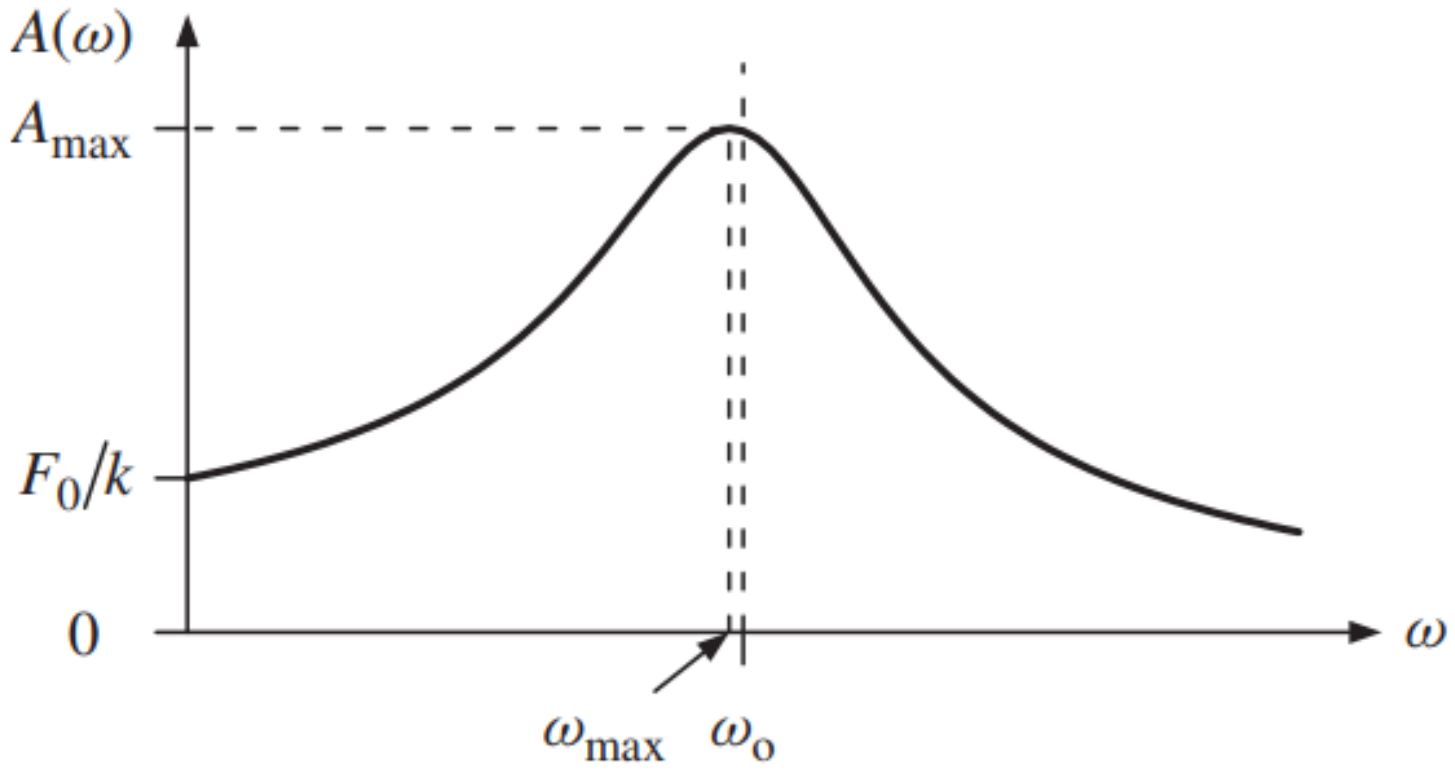
$$\sin\delta = \frac{-2\xi\omega_0\Omega}{\sqrt{4\xi^2\omega_0^2\Omega^2 + (\omega_0^2-\Omega^2)^2}}$$

$$\boxed{A^* = \frac{a/m}{\sqrt{(\omega_0^2 - \Omega^2)^2 + 4\xi^2\Omega^2\omega_0^2}}}$$

$$x(t) = \underbrace{A e^{-\xi\omega_0 t} \sin(\omega t + \varphi)}_{x_h} + \underbrace{A^* \sin(\Omega t + \delta)}_{x_p}$$

$$\xi \stackrel{\text{def}}{=} \frac{b}{b_{\text{crit}}} = \frac{b}{2m\omega_0} \quad \Rightarrow \quad \gamma = \frac{b}{m} = 2m\xi\omega_0$$

$$A(\omega) = \frac{F_0/m}{[(\omega_o^2 - \omega^2)^2 + \omega^2\gamma^2]^{1/2}}.$$



Exemplo

- Uma massa de 1,5 kg repousa sobre uma mesa horizontal e está presa a uma extremidade de uma mola de constante elástica 150 N/ m. A outra extremidade da mola é movida na direção horizontal de acordo com *x = acos ωt* onde a =5 × 10^(−3)m e *ω = 6π rad/s*. A constante de amortecimento *b = 3,0 N*s/m*. Determine a amplitude e a fase das oscilações de estado estacionário da massa.

$$\omega_o = \sqrt{k/m} = \sqrt{150/1.5} = 10 \text{ rad s}^{-1}$$

$$\gamma = b/m = 2.0 \text{ s}^{-1}.$$

$$A(\omega) = \frac{a\omega_o^2}{[(\omega_o^2 - \omega^2)^2 + \omega^2\gamma^2]^{1/2}}$$

$$A(\omega) = \frac{ak/m}{[(\omega_o^2 - \omega^2)^2 + \omega^2\gamma^2]^{1/2}}$$

$$= \frac{150 \times 5 \times 10^{-3}}{1.5\{[10^2 - (6\pi)^2]^2 + 2^2(6\pi)^2\}^{1/2}} = 1.9 \times 10^{-3} \text{ m.}$$

$$\tan\delta = \frac{\omega\gamma}{(\omega_o^2 - \omega^2)} = \frac{6\pi \times 2}{(10^2 - (6\pi)^2)} = -0.15.$$